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Ivgi(10) **Pub. No.: US 2025/0271237 A1**(43) **Pub. Date: Aug. 28, 2025**(54) **HANDGUN HOLSTER WITH RETENTION MECHANISM**(52) **U.S. Cl.**CPC *F41C 33/0227* (2013.01)(71) Applicant: **Orpaz Defense Ltd.**, Rosh HaAyin (IL)(72) Inventor: **Simon Ivgi**, Herzliya (IL)(73) Assignee: **Orpaz Defense Ltd.**, Rosh HaAyin (IL)(21) Appl. No.: **19/063,639**(22) Filed: **Feb. 26, 2025**(30) **Foreign Application Priority Data**

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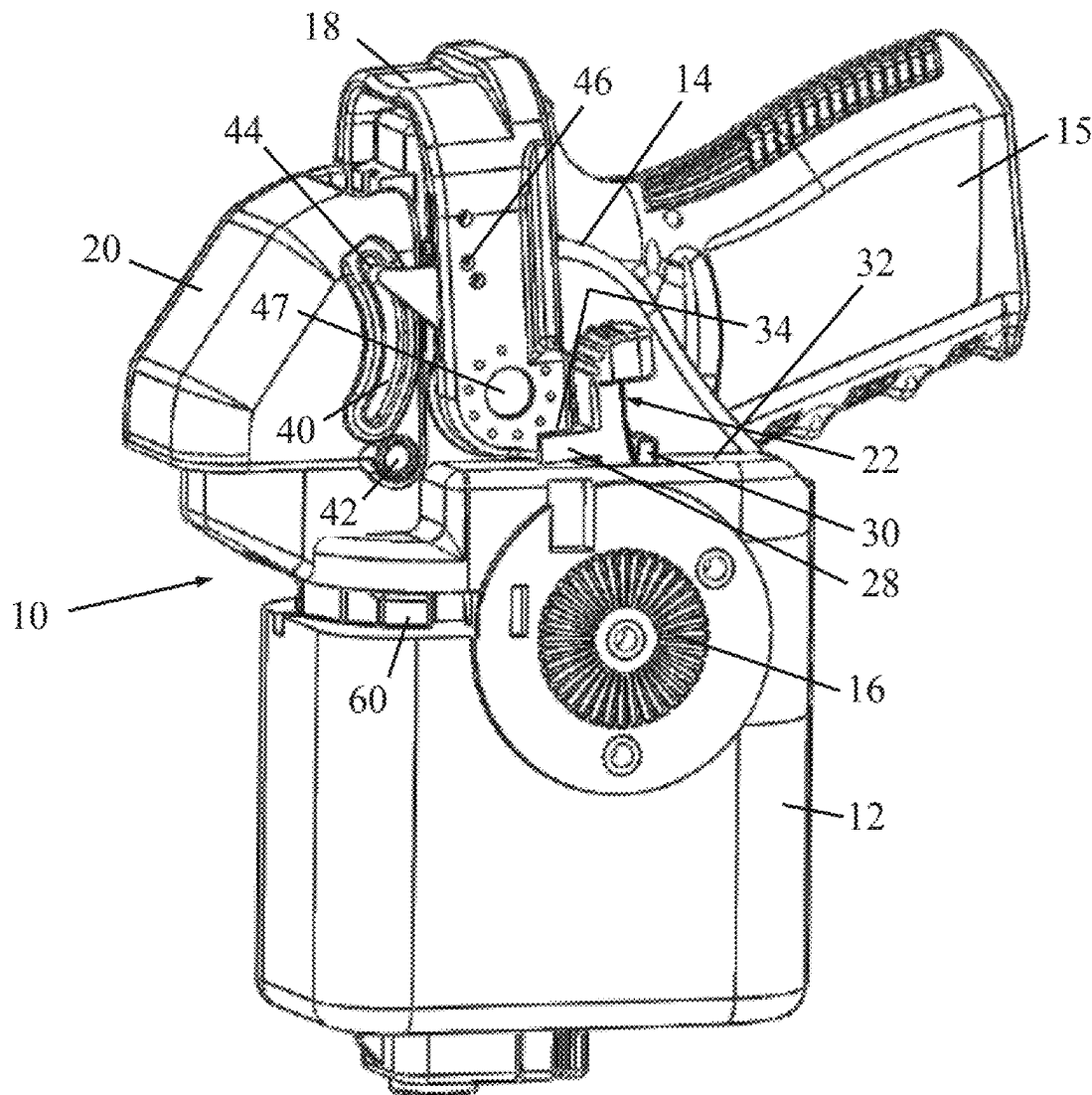
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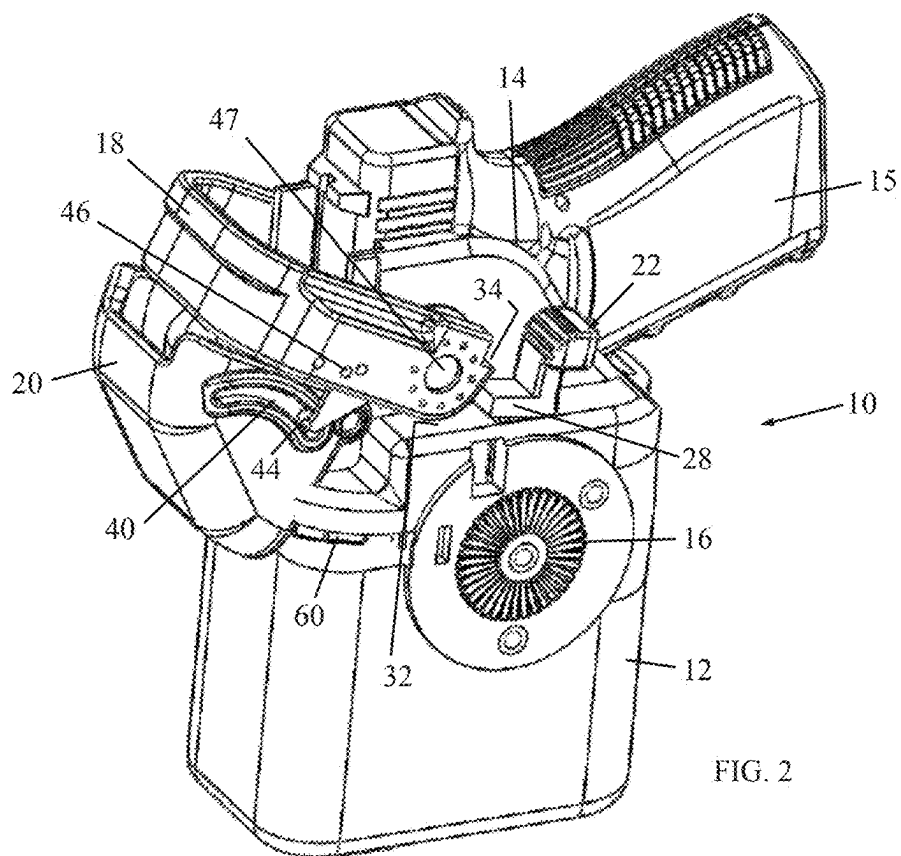
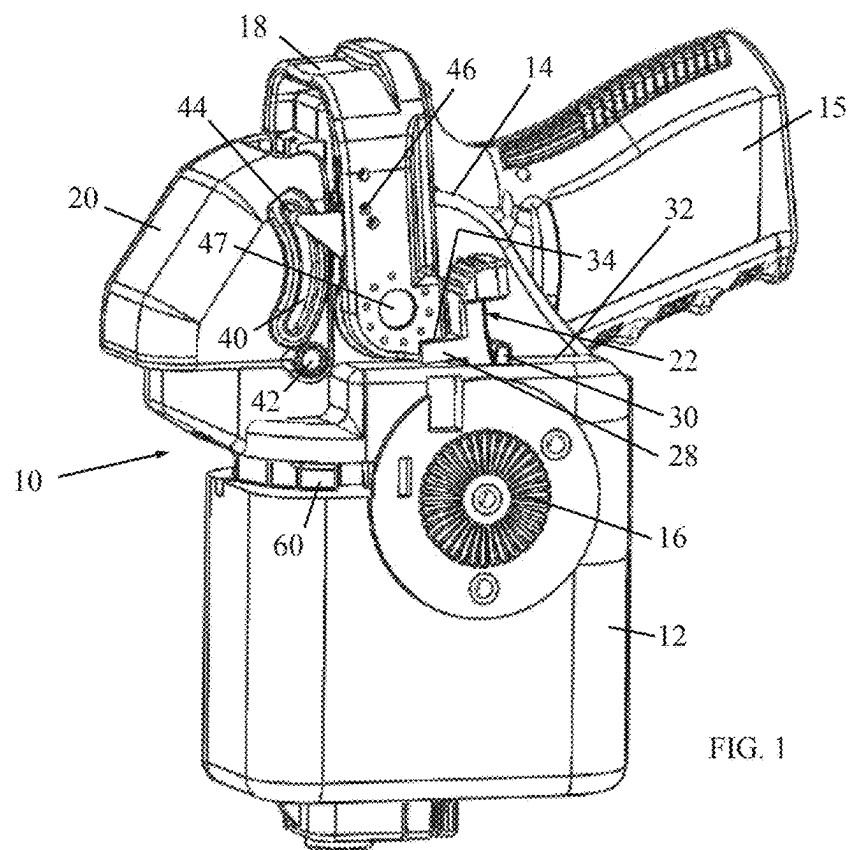
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ABSTRACT

A holster includes a retention catch arranged to enter a portion of the weapon to be in a locked position. A release lever is coupled to the holster housing, which includes a shaft arranged to push against a catch interface member that can move the retention catch out of the locked position. The release lever includes a tab which in an initial position rests on a first surface in the holster housing. The release lever is movable so that the tab moves towards a channel formed in the holster housing and the release lever is movable to overcome a biasing force of a biasing device so that the release lever enters into the channel and pushes against the catch interface member to move the retention catch out of the locked position into an unlocked position. Without application of any force on the release lever, the biasing device urges the tab against a second surface and the release lever cannot return to the initial position.





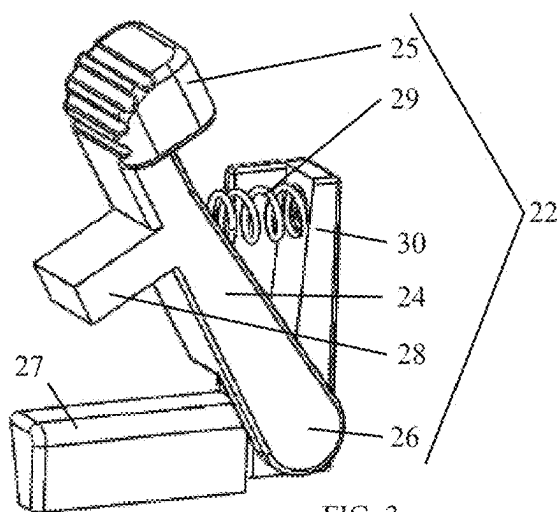


FIG. 3

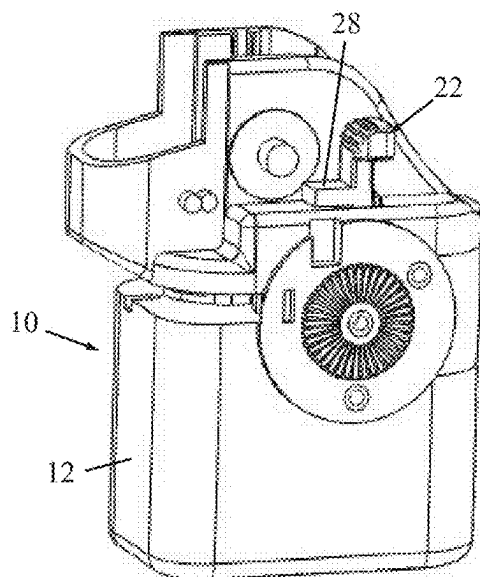


FIG. 4A

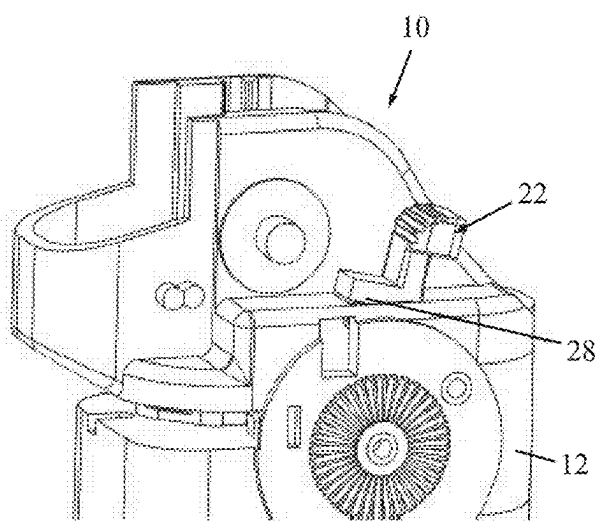


FIG. 4B

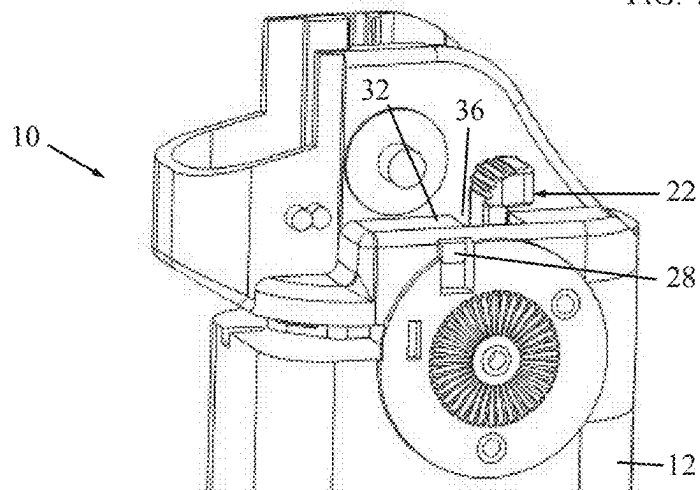
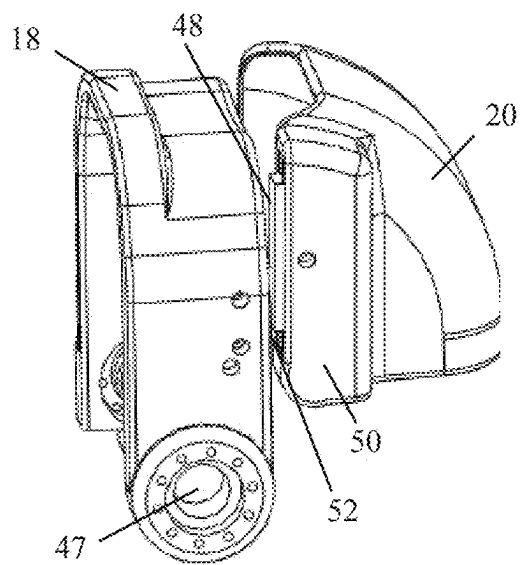
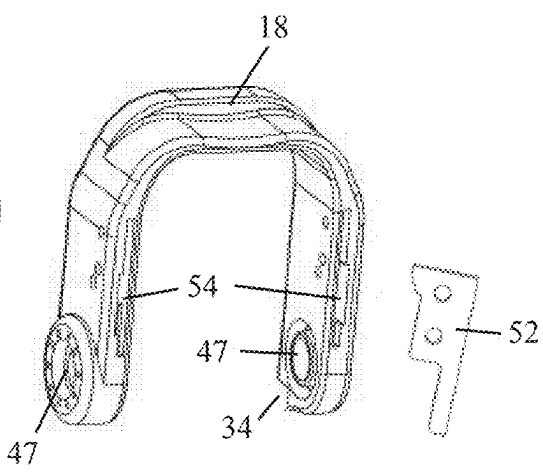
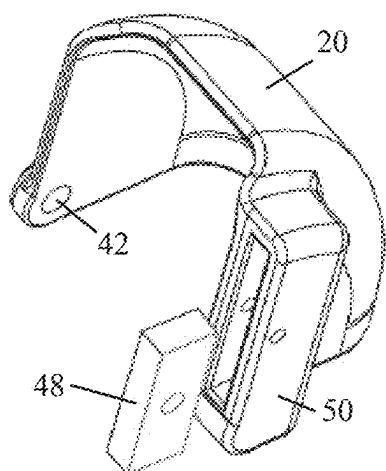
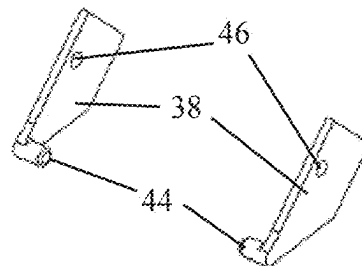
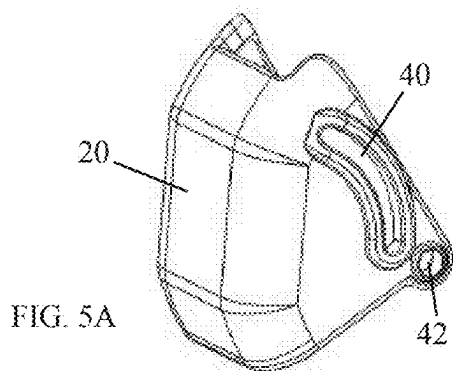
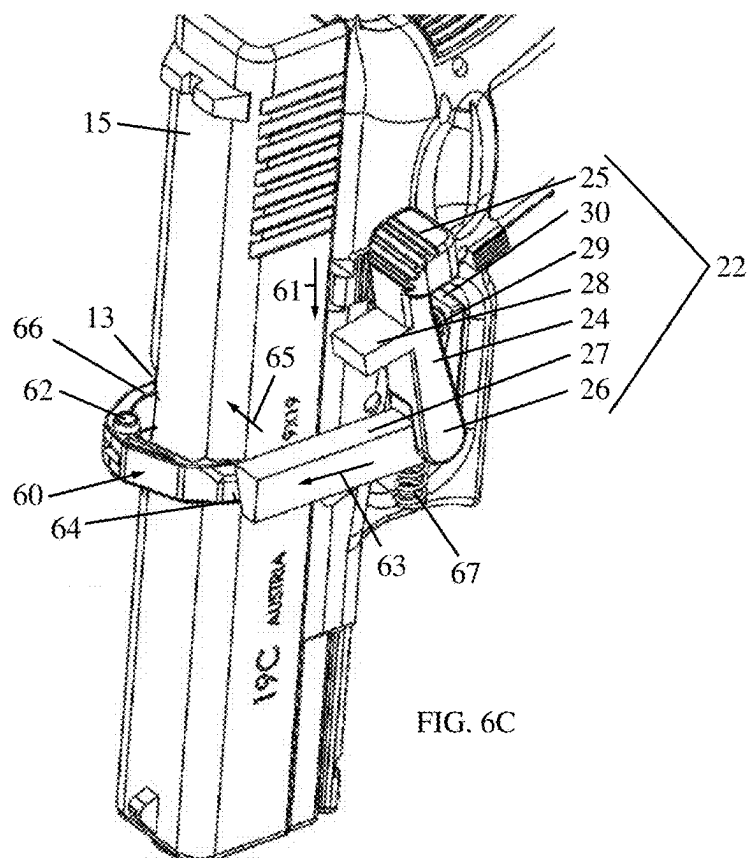
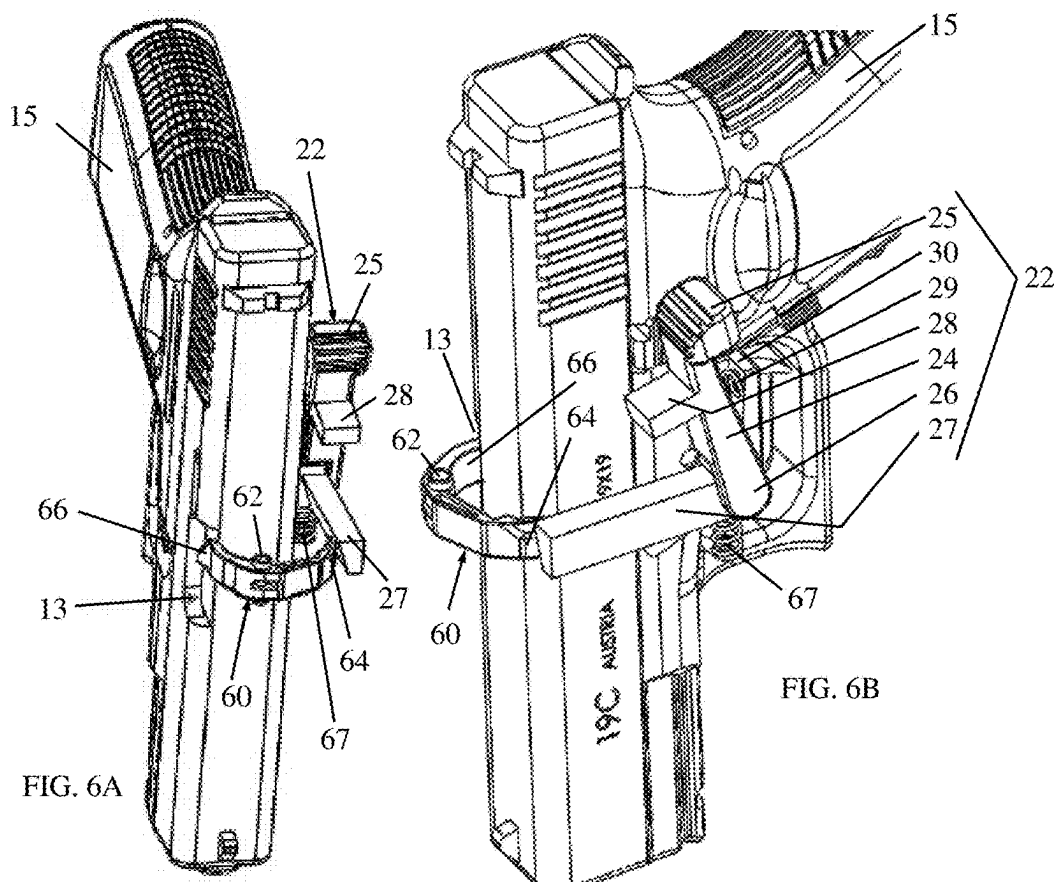


FIG. 4C





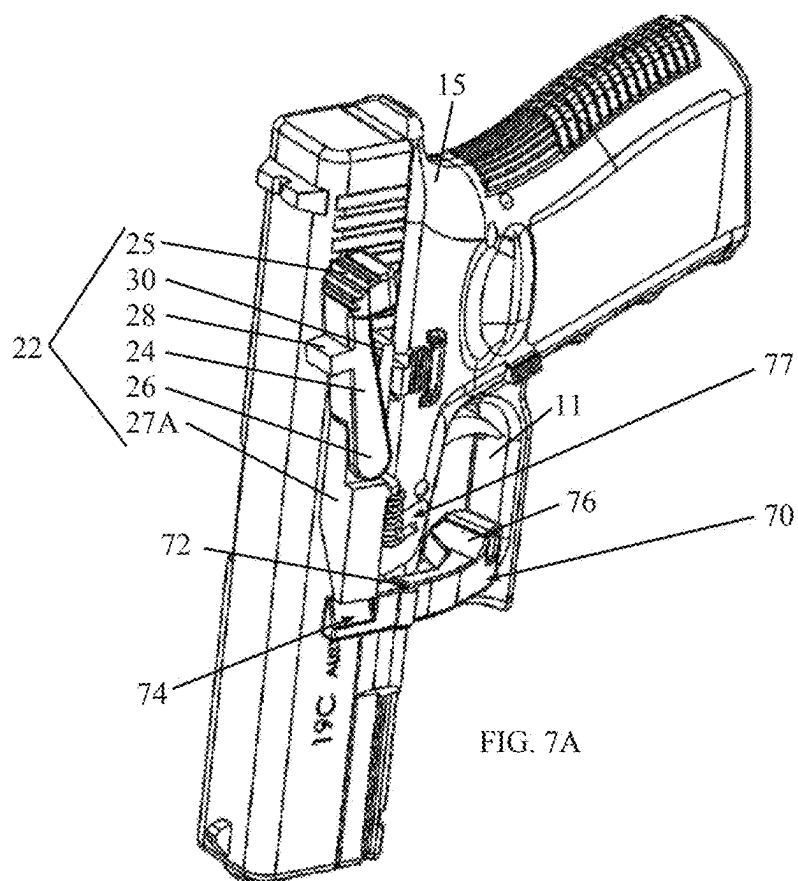


FIG. 7A

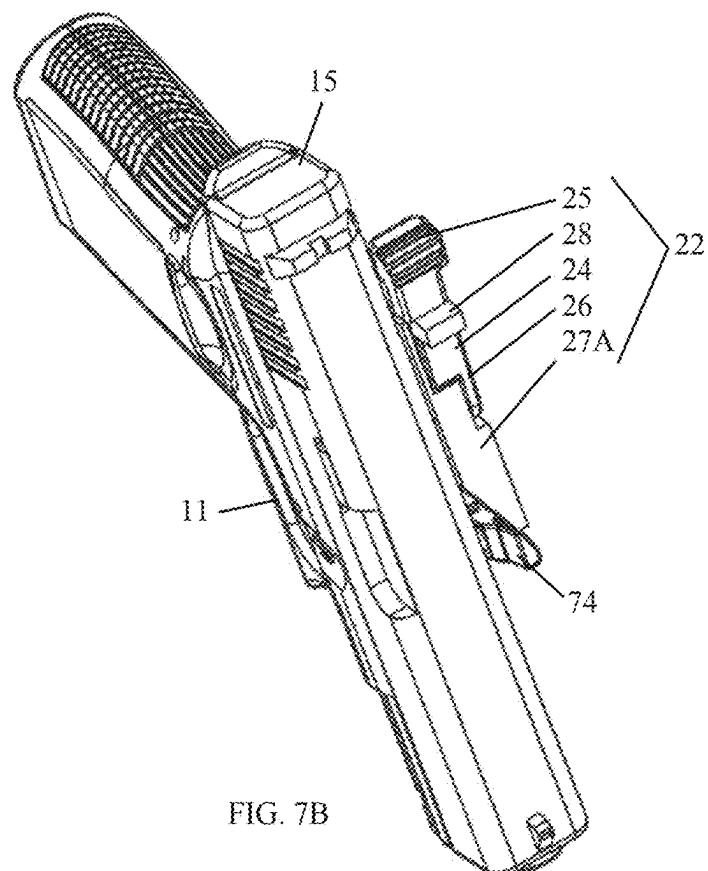
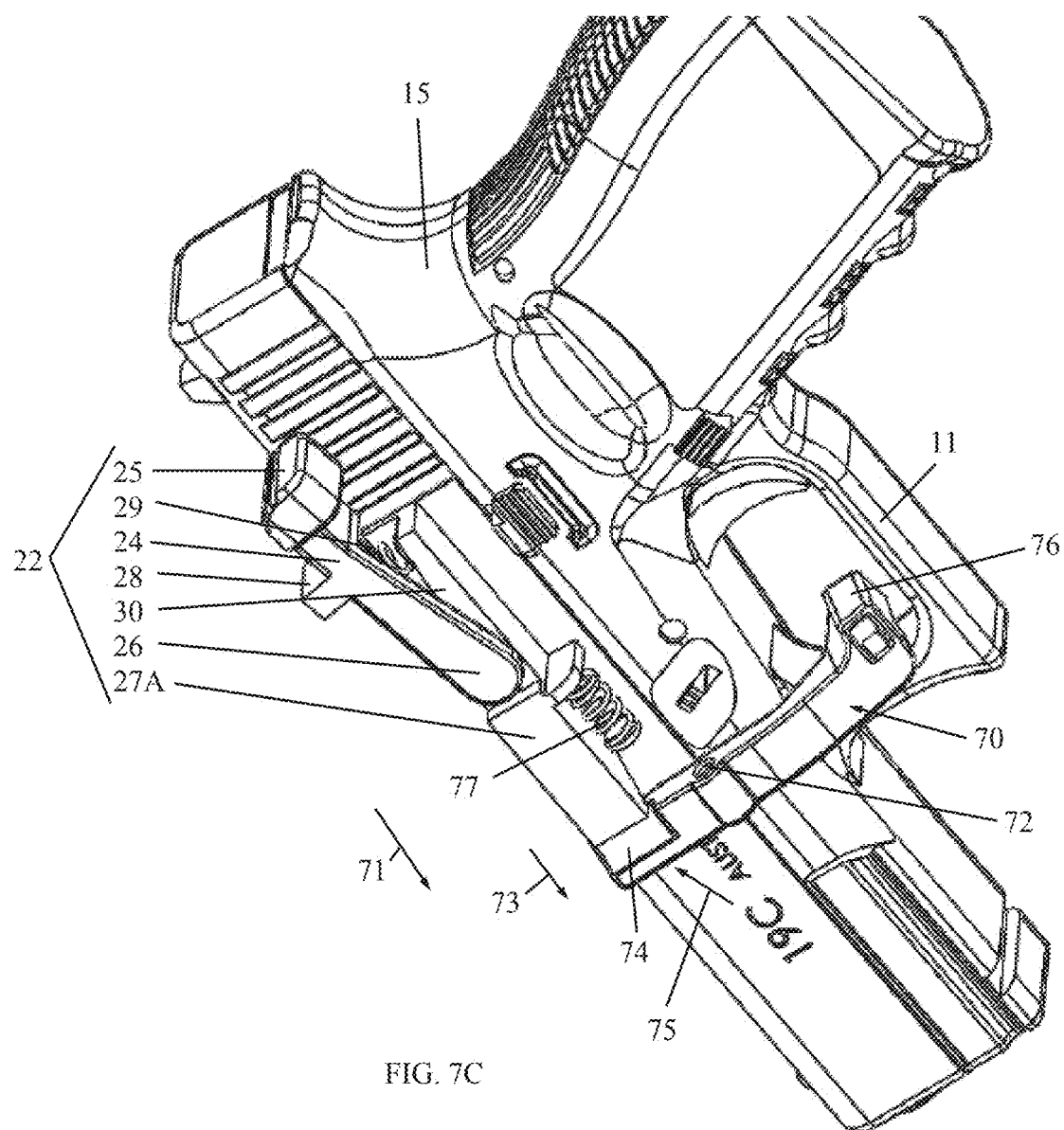
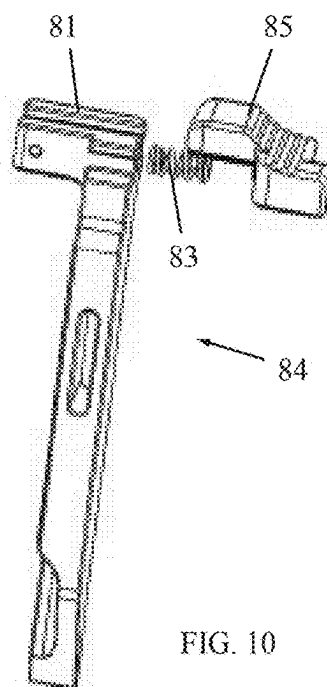
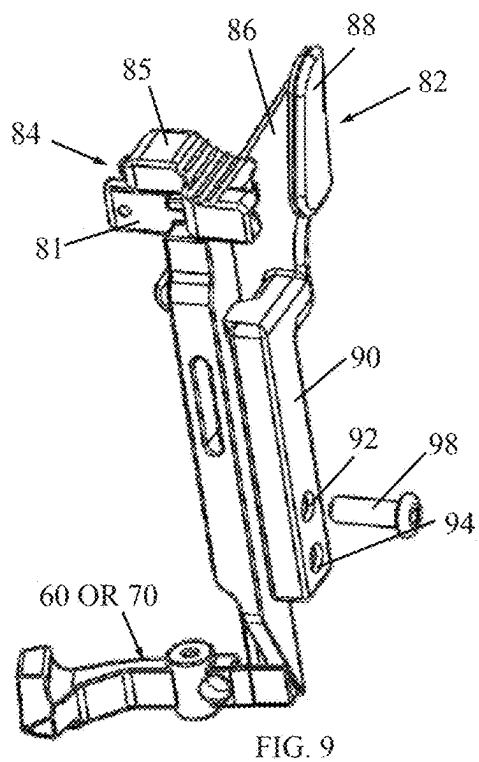
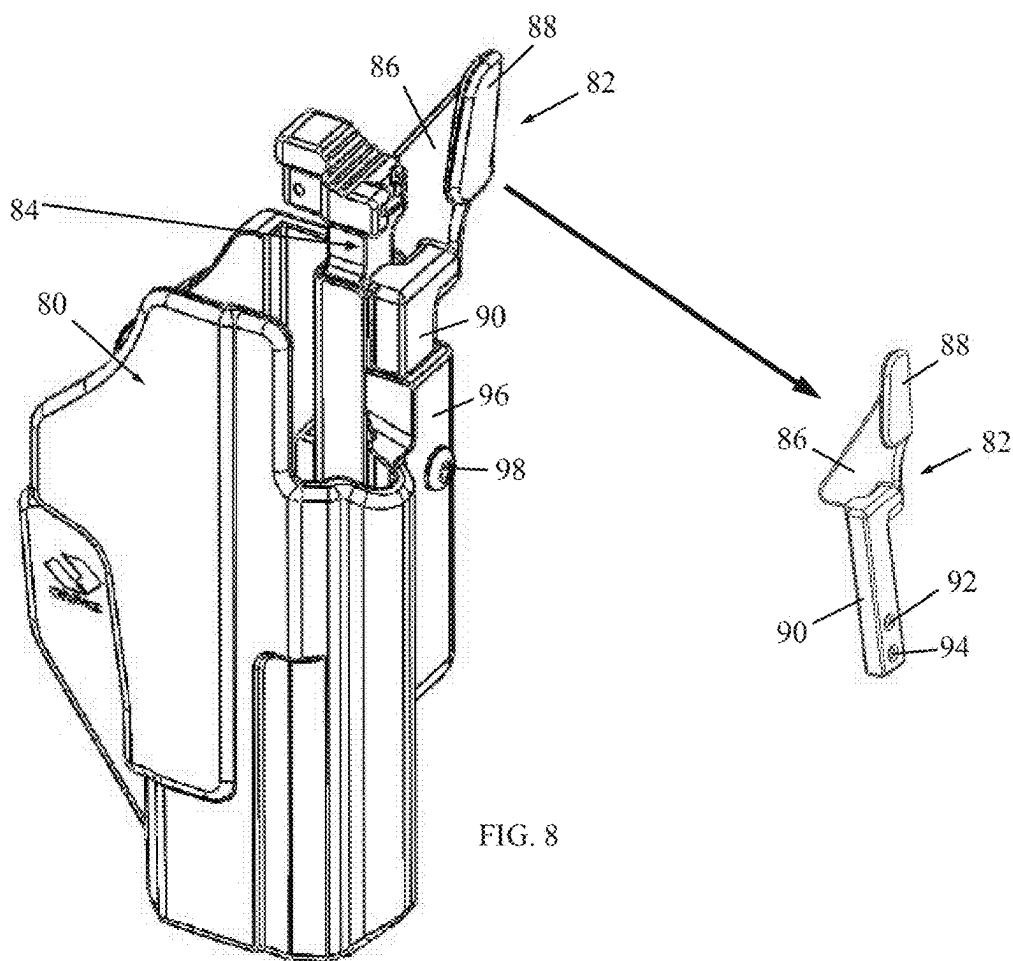


FIG. 7B





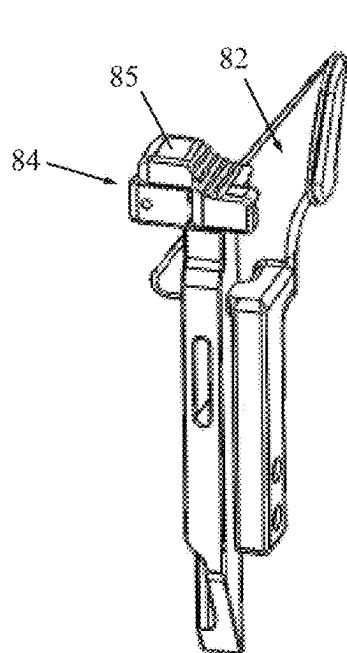


FIG. 11A

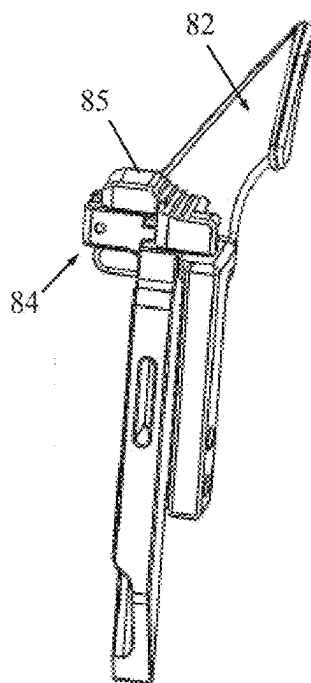


FIG. 11B

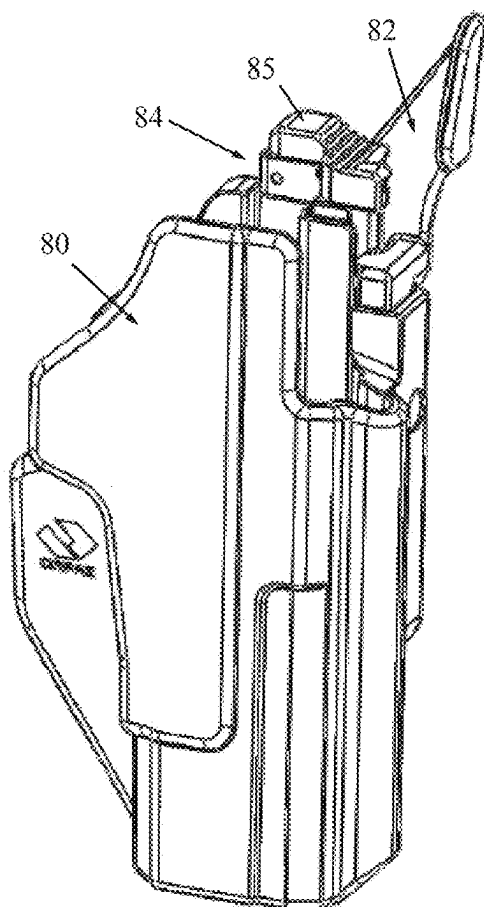


FIG. 12A

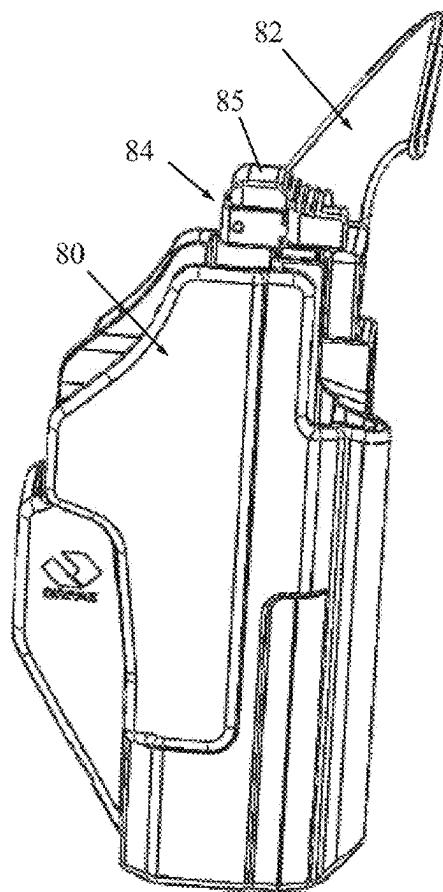


FIG. 12B

HANDGUN HOLSTER WITH RETENTION MECHANISM

FIELD OF THE INVENTION

[0001] The present invention relates to handgun holsters with a retention mechanism.

BACKGROUND OF THE INVENTION

[0002] Many users of handguns, particularly military and law enforcement personnel, carry a handgun in a holster designed to protect the handgun and hold it securely. Holsters can be worn in a number of ways, such as on a belt at the waist, on the thigh, under an arm, or around an ankle.

[0003] Some holsters include a variety of strap or flap arrangements that prevent the removal of the firearm from the holster while the strap or flap is in place. The user must first unfasten and/or rotate the strap/flap before the firearm can be withdrawn. To re-secure the handgun in the holster once the handgun has been re-holstered, the user must physically refasten and/or rotate the strap/flap before the firearm is securely retained within the holster. Some users might not prefer these designs because of the time required to release and/or re-secure the handgun.

[0004] Accordingly, a user-friendly holster retention mechanism is needed.

SUMMARY

[0005] The present invention seeks to provide user-friendly holster retention mechanism, as is described more in detail hereinbelow.

[0006] There is thus provided in accordance with a non-limiting embodiment of the invention a holster including a holster housing which has an opening for inserting there-through a weapon, and including a retention catch arranged to enter a portion of the weapon to be in a locked position, and a release lever coupled to the holster housing, the release lever including a shaft arranged to push against a catch interface member, the catch interface member being configured to move the retention catch out of the locked position, wherein the release lever includes a tab which in an initial position rests on a first surface (e.g., a first side of a shelf surface) in the holster housing, the release lever being movable so that the tab moves towards a channel formed in the holster housing and the release lever is movable to overcome a biasing force of a biasing device so that the release lever enters into the channel and pushes against the catch interface member to move the retention catch out of the locked position into an unlocked position, and in the unlocked position, without application of any force on the release lever, the biasing device urges the tab against a second surface (e.g., a second side of the shelf surface or some other surface in the holster housing) and the release lever cannot return to the initial position.

[0007] In accordance with a non-limiting embodiment of the invention a hood is arranged to pivot over a portion of the weapon, wherein in the initial position, the release lever prevents movement of the hood and wherein movement of the tab towards the channel releases and allows movement of the hood.

[0008] In accordance with a non-limiting embodiment of the invention an optics cover is arranged to pivot over a portion of the weapon, wherein in the initial position, the release lever prevents movement of the optics cover and

wherein movement of the tab towards the channel releases and allows movement of the optics cover.

[0009] In accordance with a non-limiting embodiment of the invention a hood is arranged to pivot over a first portion of the weapon and an optics cover is arranged to pivot over a second portion of the weapon, wherein in the initial position, the release lever prevents movement of the hood and the optics cover and wherein movement of the tab towards the channel releases and allows movement of the hood and the optics cover. The hood may be coupled to the optics cover so that the hood and the optics cover move together. The hood may be coupled to the optics cover by means of link arms formed with pivot axles that move in arcuate tracks formed in the optics cover. Alternatively, the hood may be coupled to the optics cover by magnetic attraction.

[0010] In accordance with a non-limiting embodiment of the invention the hood is formed with a recess and the tab is received in the recess in the initial position.

[0011] In accordance with a non-limiting embodiment of the invention the portion of the weapon in which the retention catch is arranged to enter is an ejection port or a trigger guard.

[0012] There is provided in accordance with a non-limiting embodiment of the invention a holster including a holster housing which has an opening for inserting there-through a weapon, and including a retention catch arranged to enter a portion of the weapon to be in a locked position, a release lever coupled to the holster housing, the release lever including a shaft arranged to push against a catch interface member, the catch interface member being configured to move the retention catch out of the locked position, and a retention-level member that has two mounting positions in the holster housing, the retention-level member cooperating with the release lever so that at one of the mounting positions the release lever has level-one retention and at the other of the mounting positions the release lever has level-two retention.

[0013] In accordance with a non-limiting embodiment of the invention the release lever includes a movable pusher member arranged to move against a biasing force of a biasing device.

BRIEF DESCRIPTION OF DRAWINGS

[0014] The present invention will be understood and appreciated more fully from the following detailed description, taken in conjunction with the drawings in which:

[0015] FIG. 1 is a simplified perspective illustration of a holster, in accordance with a non-limiting embodiment of the present invention, in which a release lever is in a retaining position that retains a hood and optics cover in the locked position, and is also in a retaining position that retains a retention catch in a locked position in either an ejection port or trigger guard of a weapon.

[0016] FIG. 2 is a simplified perspective illustration of the holster, in which the release lever has been moved out of the hood and optics cover retaining position so that the hood and optics cover move together out of the locked position into an unlocked position, but the release lever is still in the retaining position that retains the retention catch in the locked position in the ejection port or trigger guard of the weapon.

[0017] FIG. 3 is a simplified perspective illustration of the release lever assembly.

[0018] FIGS. 4A, 4B and 4C are perspective illustrations of three different positions of the release lever assembly, in which FIG. 4A is the totally locked position in which the hood and optics cover are in the locked position and the retention catch for the ejection port or trigger guard is in the locked position, FIG. 4B is the semi-locked position in which the hood and optics cover are unlocked but the retention catch remains locked (this first movement from totally locked to semi-locked, which is a horizontal and/or a pivoting movement, is level-one retention), and FIG. 4C is the fully unlocked position in which the hood and optics cover and the retention catch are all unlocked (this second movement from semi-locked to fully unlocked, which is a vertical movement, is level-two retention). FIG. 4C shows the resting position of the release lever after having been moved to the fully unlocked position, showing that the release lever cannot move back to the semi-locked position (and of course not back to the fully locked position) unless moved by the user horizontally to clear the obstacle that prevents upward movement of the release lever.

[0019] FIGS. 5A and 5B are respectively simplified perspective illustrations of the optics cover and link arms that couple the hood with arcuate tracks in the optics cover so that the optics cover moves with the hood.

[0020] FIGS. 5C and 5D are respectively simplified perspective illustrations of the optics cover with magnets and magnetic elements in the hood that are magnetically attracted to each other so that the optics cover moves with the hood.

[0021] FIG. 5E is a simplified perspective illustration of the optics cover magnetically attracted to the hood.

[0022] FIGS. 6A and 6B are simplified front and side perspective illustrations, respectively, of the release lever engaged with the retention catch which is locked in the ejection port of the weapon.

[0023] FIG. 6C is a simplified perspective illustration of the release lever about to push against a portion of the retention catch mechanism in order to pivot the retention catch out of the locked engagement in the ejection port of the weapon.

[0024] FIGS. 7A and 7B are simplified side and front perspective illustrations, respectively, of the release lever engaged with the retention catch which is locked in the trigger guard of the weapon.

[0025] FIG. 7C is a simplified perspective illustration of the release lever about to push against a portion of the retention catch mechanism in order to pivot the retention catch out of the locked engagement in the trigger guard of the weapon.

[0026] FIG. 8 is a simplified perspective illustration of a holster, in accordance with a non-limiting embodiment of the present invention, in which a retention-level member cooperates with a release lever so that the user can select if the release lever has level-one or level-two retention.

[0027] FIG. 9 is a simplified perspective illustration of the retention-level member cooperates and the release lever, showing that the retention-level member has two assembly positions, one for level-one retention and the other for level-two retention.

[0028] FIG. 10 is an exploded illustration of the release lever showing a pushing element biased by a biasing device, such as a coil spring.

[0029] FIGS. 11A and 11B are perspective illustrations, respectively, of the retention-level member mounted at the

level-one retention position, in which the release lever has to be moved in one movement (vertically) to release any catch holding the weapon, and the retention-level member mounted at the level-two retention position, in which the release lever has to be moved in two movements (horizontally and then vertically) to release any catch holding the weapon.

[0030] FIGS. 12A and 12B are perspective illustrations, respectively, of the holster with the retention-level member mounted at the level-one retention position, in which the release lever has to be moved in one movement (vertically) to release any catch holding the weapon, and the retention-level member mounted at the level-two retention position, in which the release lever has to be moved in two movements (horizontally and then vertically) to release any catch holding the weapon.

DETAILED DESCRIPTION

[0031] Reference is now made to FIGS. 1 and 2, which illustrate a holster 10, in accordance with a non-limiting embodiment of the present invention.

[0032] Holster 10 may include a holster housing 12, made of any suitable holster material, such as but not limited to, a polymer, such as KYDEX (thermoplastic acrylic-polyvinyl chloride). The holster housing 12 has an opening 14 for inserting therethrough a weapon 15. The holster housing 12 may include a paddle hub 16 (which may be serrated), or any other garment attachment means, as is known in the art.

[0033] Holster may include a hood 18 and an optics cover 20. As seen in FIG. 1, when in the locked position, hood 18 lies over a rear portion of the weapon 15, such as but not limited to, the rear face of the slide, and optics cover 20 lies over a portion of the weapon where an optics accessory may be mounted, such as but not limited to, a laser aiming accessory mounted on a Picatinny rail on the upper side of the slide. As will be described below, hood 18 and optics cover 20 move together and are actuated by a release lever 22.

[0034] Reference is now made to FIG. 3, which illustrates the assembly of the release lever 22. Release lever 22 may include a shaft 24 which has a pusher member 25 at an upper end thereof. The terms "upper" and "lower" are non-limiting relative terms, and refer to the positions shown in the drawings. A lower end 26 of shaft 24 is arranged to push against a catch interface member 27. As will be described below, the catch interface member 27 is configured to move a retention catch 60 (seen partially in FIGS. 1 and 2) out of a locked position in either an ejection port or trigger guard (catch 70, FIG. 7A) of the weapon.

[0035] A tab 28 protrudes sideways from shaft 24 at a position intermediate pusher member 25 and lower end 26 of shaft 24. A biasing device 29, such as a coil spring, provides a biasing force against shaft 24. Biasing device 29 is placed against a counterforce surface 30. The biasing force urges release lever 22 to be in the locked position, until the release lever 22 is moved by the user to an unlocked position.

[0036] Reference is now made to FIGS. 4A, 4B and 4C are three different positions of the release lever 22. In FIG. 4A, the release lever 22 is in the totally locked position in which tab 28 lies over a shelf 32 of holster housing 12. This is the position shown in FIG. 1. As seen in FIG. 1, in this totally locked position, hood 18 and optics cover 20 are in the locked position. As will be explained below, the retention catch for the ejection port or trigger guard is also in the

locked position. As seen in FIGS. 1 and 2, hood 18 may be formed with a recess 34. In the locked position of FIG. 1, tab 28 is received in recess 34 and thus prevents hood 18 (and optics cover 20) from moving out of the locked position.

[0037] In FIG. 4B, release lever 22 has been moved to the right (in the sense of the drawings) which moves tab 28 out of recess 34 of hood 18 (as seen in FIG. 2). This is the semi-locked position in which the hood 18 and optics cover 20 are unlocked. This first movement is a movement from the totally locked position to the semi-locked position, which is a horizontal and/or a pivoting movement, and provides level-one retention. In this position, the retention catch remains locked, as explained below.

[0038] FIG. 4C is the fully unlocked position in which the hood and optics cover and the retention catch are all unlocked. This is accomplished by moving the release lever 22 vertically downwards (in the sense of the drawings) into a channel 36 formed in holster housing 12. This second movement from the semi-locked position to the fully unlocked position provides level-two retention. FIG. 4C shows the resting position of the release lever 22 after having been moved to the fully unlocked position, showing that the release lever 22 cannot move back to the semi-locked position (and of course not back to the fully locked position) unless moved by the user horizontally to clear the underside of shelf 32, which is the obstacle that prevents upward movement of the release lever 22.

[0039] Reference is now made to FIGS. 5A and 5B, which illustrate the optics cover 20 and link arms 38 that couple the hood 18 (not shown here, but shown in FIGS. 1 and 2). Optics cover 20 may be formed with arcuate tracks 40 and a pivot 42. Each of the link arms 38 may be formed with a pivot axle 44, and with a mounting provision 46, such as a mounting hole.

[0040] As seen in FIGS. 1 and 2, the pivot axle 44 slides in arcuate track 40. Hood 18 pivots about a hood pivot 47. Moving release lever from the locked position of FIG. 1 to the unlocked position of FIG. 2 allows hood 18 and optics cover 20 to move together and pivot downwards to the unlocked position. A biasing device (not shown), such as a coil spring at the pivot 32 or 47 or other suitable position, may urge the hood 18 and optics cover 20 to move to the unlocked position when release lever 22 is moved out of the way. The user may manually move hood 18 and optics cover 20 back to their original positions, by simply pushing them back to the original positions against the force of the biasing device that urged them to the open positions.

[0041] It is noted that the holster can be provided with hood 18 and without the optics cover 20, or with the optics cover 20 and without the hood 18.

[0042] Reference is now made to FIGS. 5C and 5D, which show an alternative way of hood 18 moving together with optics cover 20. In this variation, as seen in FIG. 5C, optics cover 20 may be provided with a magnet 48, which may be made of any suitable magnetic material, such as but not limited to, rare-earth materials, e.g., neodymium iron boron or samarium cobalt and the like, or non-rare-earth materials, e.g., different ferrous alloys. The magnet 48 may be mounted in an insert 50.

[0043] As seen in FIG. 5D, hood 18 may be provided with a magnetic element 52 mounted in a crevice 54 formed in hood 18. The magnetic element 52 may be made of a magnetic ferrous alloy, such as various steel alloys. As seen in FIG. 5E, the magnet 48 of optics cover 20 and the

magnetic element 52 of hood 18 are magnetically attracted to each other so that the optics cover 20 moves with the hood 18.

[0044] Reference is now made to FIGS. 6A and 6B, which illustrate the release lever 22 engaged with the retention catch 60 which is locked in an ejection port 13 of weapon 15. The retention catch 60 may be shaped as an arcuate lever that pivots about a lever pivot 62. One end of retention catch 60 is an inclined interface surface 64 that interfaces with catch interface member 27 of release lever 22. The opposite end of retention catch 60 is a tongue 66 that enters the ejection port 13.

[0045] Reference is now made to FIG. 6C. Release lever 22 may be moved downwards (arrow 61) which causes catch interface member 27 to move sideways (arrow 63) on inclined interface surface 64. The downward movement is possible only when release lever has been moved to the position of FIG. 4B and 4C when it can then be pushed downwards in channel 36 (FIG. 4C). The downward movement causes inclined interface surface 64 to move inwards (arrow 65) and pivot the tongue 66 of retention catch 60 out of the locked engagement in the ejection port 13 of the weapon 15. A biasing device 67, such as a coil spring, urges catch interface member 27 upwards against release lever 22 so as to normally maintain tongue 66 of retention catch 60 in the locked engagement in ejection port 13.

[0046] Reference is now made to FIGS. 7A and 7B, which illustrate the release lever 22 engaged with a retention catch 70 which is locked in a trigger guard 11 of weapon 15. The retention catch 70 may be shaped as an arcuate lever that pivots about a lever pivot 72. One end of retention catch 70 is an inclined interface surface 74 that interfaces with a catch interface member 27A of release lever 22. The opposite end of retention catch 70 is a tongue 76 that enters the trigger guard 11.

[0047] Reference is now made to FIG. 7C. Release lever 22 may be moved downwards (arrow 71) which causes catch interface member 27A to move downwards (arrow 73) on inclined interface surface 74. Again, the downward movement is possible only when release lever has been moved to the position of FIG. 4B and 4C when it can then be pushed downwards in channel 36 (FIG. 4C). The downward movement causes inclined interface surface 74 to move inwards (arrow 75) and pivot the tongue 76 of retention catch 70 out of the locked engagement in the trigger guard 11 of the weapon 15. A biasing device 77, such as a coil spring, urges catch interface member 27A upwards against release lever 22 so as to normally maintain tongue 76 of retention catch 70 in the locked engagement in trigger guard 11.

[0048] Reference is now made to FIG. 8, which illustrates a holster 80, in accordance with a non-limiting embodiment of the present invention. Holster 80 may include a holster housing made similar to holster 10. Holster 80 includes a retention-level member 82 that cooperates with a release lever 84 so that the user can select if the release lever 84 has level-one or level-two retention, as is now described.

[0049] Reference is now made additionally to FIG. 9. The retention-level member 82 may include an elongate member 86 that has a grasping member 88 at one portion thereof and a position selection member 90 at another portion thereof. The position selection member 90 may be formed with two mounting holes 92 and 94 (FIG. 9). As seen in FIG. 8, position selection member 90 may be received in a channel housing 96 in holster 80. A fastener 98, such as a screw, may

secure position selection member **90** to channel housing **96** at either mounting hole **92** or **94**.

[0050] As seen in FIG. 9, release lever **84** may be arranged to operate retention catch **60** or **70**, as described above. Release lever **84** differs from release lever **22** in that release lever **84** has a movable pusher member **85** which slides along a head **81** and is biased by a biasing device **83** (FIG. 10).

[0051] Reference is now made to FIGS. 11A and 12A, which illustrate the retention-level member **82** mounted at the level-one retention position, in which the release lever **84** has to be moved in one movement (vertically downwards) to release any catch holding the weapon.

[0052] Reference is now made to FIGS. 11B and 12B, which illustrate the retention-level member **82** mounted at the level-two retention position, in which the release lever **84** has to be moved in two movements (horizontally and then vertically) to release any catch holding the weapon. In this position, movable pusher member **85** is blocked from downwards movement because it abuts against the top of position selection member **90**. The movable pusher member **85** must first be moved horizontally against the force of the biasing device **83** (FIG. 10) (this is the first movement) and only then the release lever **84** can be moved downwards (the second movement) to release any catch holding the weapon.

[0053] It is noted that retention-level member **82** and release lever **84** may be incorporated in the embodiment of holster **10**. If such an embodiment is used with the hood **18**, when the retention-level member **82** is mounted for level-two retention of the release lever **84**, the addition of hood **18** makes the holster have a total of level-three retention. This is because the hood **18** also protects the handgun from being removed from the holster and the hood **18** cannot be released unless the user first moves movable pusher member **85** horizontally against the force of the biasing device **83** (FIG. 10). In such an embodiment, after the user moves movable pusher member **85** horizontally, then release lever **84** can be moved down in the channel **36** (FIG. 4C) to release hood **18** (and optics cover **20**, if provided, which moves together with hood **18**) and to release the catches in the ejection port or trigger guard.

1. A holster comprising:

- a holster housing which has an opening for inserting therethrough a weapon, and comprising a retention catch arranged to enter a portion of the weapon to be in a locked position; and
- a release lever coupled to said holster housing, said release lever comprising a shaft arranged to push against a catch interface member, said catch interface member being configured to move said retention catch out of said locked position;

wherein said release lever comprises a tab which in an initial position rests on a first surface in said holster housing, said release lever being movable so that said tab moves towards a channel formed in said holster housing and said release lever is movable to overcome a biasing force of a biasing device so that said release lever enters into said channel and pushes against said catch interface member to move said retention catch out of said locked position into an unlocked position, and in said unlocked position, without application of any force on said release lever, said biasing device urges said tab against a second surface and said release lever cannot return to said initial position.

2. The holster according to claim 1, further comprising a hood arranged to pivot over a portion of the weapon, wherein in said initial position, said release lever prevents movement of said hood and wherein movement of said tab towards said channel releases and allows movement of said hood.

3. The holster according to claim 1, further comprising an optics cover arranged to pivot over a portion of the weapon, wherein in said initial position, said release lever prevents movement of said optics cover and wherein movement of said tab towards said channel releases and allows movement of said optics cover.

4. The holster according to claim 1, further comprising a hood arranged to pivot over a first portion of the weapon and an optics cover arranged to pivot over a second portion of the weapon, wherein in said initial position, said release lever prevents movement of said hood and said optics cover and wherein movement of said tab towards said channel releases and allows movement of said hood and said optics cover.

5. The holster according to claim 4, wherein said hood is coupled to said optics cover so that said hood and said optics cover move together.

6. The holster according to claim 5, wherein said hood is coupled to said optics cover by means of link arms formed with pivot axles that move in arcuate tracks formed in said optics cover.

7. The holster according to claim 5, wherein said hood is coupled to said optics cover by magnetic attraction.

8. The holster according to claim 2, wherein said hood is formed with a recess and said tab is received in said recess in said initial position.

9. The holster according to claim 1, wherein the portion of the weapon in which said retention catch is arranged to enter is an ejection port or a trigger guard.

10. The holster according to claim 1, further comprising a retention-level member that has two mounting positions in said holster housing, said retention-level member cooperating with said release lever so that at one of said mounting positions said release lever has level-one retention and at the other of said mounting positions said release lever has level-two retention.

11. The holster according to claim 10, wherein said release lever comprises a movable pusher member arranged to move against a biasing force of a biasing device.

12. A holster comprising:

- a holster housing which has an opening for inserting therethrough a weapon, and comprising a retention catch arranged to enter a portion of the weapon to be in a locked position;
- a release lever coupled to said holster housing, said release lever comprising a shaft arranged to push against a catch interface member, said catch interface member being configured to move said retention catch out of said locked position; and

a retention-level member that has two mounting positions in said holster housing, said retention-level member cooperating with said release lever so that at one of said mounting positions said release lever has level-one retention and at the other of said mounting positions said release lever has level-two retention.

13. The holster according to claim **12**, wherein said release lever comprises a movable pusher member arranged to move against a biasing force of a biasing device.

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